

Study: Asteroid Hit Created Two ‘Grand Canyons’ on Moon

A new study suggests an asteroid strike on the moon billions of years ago left two huge valleys about the size of Earth’s Grand Canyon.

Researchers from the United States and Britain said their research shows the two steep valleys, or canyons, likely formed in less than 10 minutes.

They noted the crash, or impact, happened near the south pole on the far side of the moon. That is the side of the moon which always faces away from Earth. The area is known as the Schrödinger impact basin.

The speed of the incoming object – which the team said might have been a comet – was estimated to be around 55,000 kilometers per hour. The strike likely happened about 3.8 billion years ago.

The researchers used data collected by a spacecraft operated by the American space agency NASA. That vehicle, the Lunar Reconnaissance Orbiter, has been capturing lunar data since entering orbit around the moon in June 2009.

One main goal of the orbiter’s mission was to create a highly detailed, three-dimensional, or 3D, map of the moon. The data has been used by NASA to learn more about lunar resources and to plan for future missions. The agency noted that the Lunar Reconnaissance Orbiter had “already collected as much data as all other planetary missions combined.”

The team studying the canyons used the orbiter’s map data to simulate the path of the incoming object, as well as material that was displaced by the crash. The scientists recently published a study describing their results in Nature Communications.

The group’s simulations suggested the space rock passed over the south pole before hitting. It then created a huge basin and launched rocky materials at speeds up to 1 kilometer per second.

The researchers said material was sent flying and struck the lunar surface like missiles. This resulted in the creation of the canyons.

The team said the canyons appear comparable in size to the Grand Canyon, in the U.S. state of Arizona.

The study’s lead writer was David Kring of the Lunar and Planetary Institute in Houston. He told the Associated Press that findings suggest the asteroid strike was a “very violent” geologic process.

Kring and his team estimated the asteroid was 25 kilometers across. They estimated that the energy needed to create the two canyons would have been more than 130 times greater than the power of all the world's current nuclear weapons.

Kring said most of the rock and debris was thrown in a direction away from the south pole. That finding is good news for NASA which plans to land astronauts in that area in the future.

Kring noted it should improve the exploration possibilities of future astronauts.

"Because debris from the Schrödinger impact was jettisoned away from the lunar south pole, ancient rocks in the polar region will be at or close to the surface....," Kring told Reuters news agency. This should make it easier for future American astronauts to collect them.

NASA's future lunar exploration plans are known as the agency's Artemis program. The program aims to return astronauts to the moon for the first time since NASA's Apollo 17 mission in 1972. The agency's current plan is to send astronauts on a trip around the moon next year. That would be followed a year or so later by a landing attempt on the lunar surface.

The researchers said older rocks in the area can help scientists gain a better understanding of the moon's past history. They could also help answer questions about the formation of Earth.

Kring noted it is currently not clear whether the two huge canyons are permanently shadowed like some of the craters at the moon's south pole. "That is something that we're clearly going to be reexamining," he said.

I'm Bryan Lynn.

Bryan Lynn wrote this story for VOA Learning English, based on reports from The Associated Press, Reuters, The Universities Space Research Association and Nature Communications.

Quiz - Study: Asteroid Hit Created Two 'Grand Canyons' on Moon

Vocabulary

asteroid - n. วัตถุที่เป็นหินซึ่งโคจรรอบดวงอาทิตย์เหมือนดาวเคราะห์

comet - n. วัตถุในอวกาศที่ทิ้งแนวเส้นสว่างไว้เบื้องหลังในท้องฟ้า

three-dimensional (3D) - adj. ที่มีความสูง ความกว้าง และความยาว

simulate - v. การทำหรือสร้างบางสิ่งให้มีพฤติกรรมหรือรูปลักษณ์เหมือนของจริงแต่ไม่ใช่

basin - n. พื้นที่ต่ำบนพื้นดินซึ่งน้ำไหลลงสู่แหล่งน้ำอื่น

jettison - v. การโยนวัสดุจากที่หนึ่งไปยังอีกที่หนึ่ง

shadow - n. พื้นที่มืดที่เกิดจากการมีบางสิ่งมาบดบังแสง

Reading Comprehension Quiz

1. What is the primary discovery described in the new study?

- A) Evidence of ancient volcanic activity on the moon.
- B) The formation of two large valleys on the moon due to an asteroid strike.
- C) A detailed map of the moon's surface near the north pole.
- D) The estimated speed of a comet hitting Earth.

2. How quickly did the two steep valleys, or canyons, on the moon likely form?

- A) Over several centuries.
- B) In more than 10 minutes.
- C) In less than 10 minutes.
- D) Over a period of billions of years.

3. Where on the moon did the asteroid impact occur?

- A) Near the north pole on the side facing Earth.
- B) Near the equator on the far side of the moon.
- C) Near the south pole on the far side of the moon.
- D) In the Sea of Tranquility.

4. Approximately how long ago did the asteroid strike the moon?

- A) About 55,000 years ago.
- B) About 3.8 billion years ago.
- C) In June 2009.
- D) In 1972.

5. Which American space agency operates the spacecraft that collected data for this study?

- A) ESA
- B) Roscosmos
- C) NASA
- D) JAXA

6. What was one of the main goals of the Lunar Reconnaissance Orbiter's mission?

- A) To search for alien life forms.
- B) To collect samples of lunar soil directly.
- C) To create a highly detailed, 3D map of the moon.
- D) To test new propulsion technologies.

7. What remarkable achievement is attributed to the Lunar Reconnaissance Orbiter regarding data collection?

- A) It was the first orbiter to reach the moon.
- B) It collected data for only one year.
- C) It collected as much data as all other planetary missions combined.
- D) Its data was only used for mapping Earth.

8. Where did the scientists recently publish their study results?

- A) The Associated Press
- B) Reuters
- C) Nature Communications
- D) The Lunar and Planetary Institute

9. According to the group's simulations, what was the path of the space rock *before* it hit the moon?

- A) It approached from the side facing Earth.
- B) It orbited the moon several times.
- C) It passed over the south pole.
- D) It approached directly from the sun.

10. What was the direct cause of the creation of the canyons, according to the researchers?

- A) Tectonic plate movement on the moon.
- B) Ancient lunar rivers eroding the surface.
- C) Material sent flying from the impact striking the lunar surface like missiles.
- D) Gradual weathering by solar winds over billions of years.

11. To what famous Earth feature are the lunar canyons compared in size?

- A) The Mariana Trench.
- B) Mount Everest.
- C) The Grand Canyon.
- D) The Sahara Desert.

12. What was the estimated energy released by the asteroid strike compared to nuclear weapons?

- A) Equal to the power of all current nuclear weapons.
- B) Less than the power of a single nuclear weapon.

- C) More than 130 times greater than the power of all the world's current nuclear weapons.
- D) Exactly double the power of all current nuclear weapons.

13. Why is the finding that debris was thrown away from the south pole considered good news for NASA?

- A) It means there will be less light pollution in the area.
- B) NASA plans to land astronauts in that area in the future.
- C) It confirmed the existence of water ice at the pole.
- D) It makes the moon's south pole a suitable location for a new observatory.

14. According to Kring, how will the direction of the jettisoned debris benefit future American astronauts?

- A) It will create new landing sites for spacecraft.
- B) It will make it easier to collect ancient rocks from the polar region.
- C) It will help them navigate the canyons more easily.
- D) It will provide protection from solar radiation.

15. What is the main objective of NASA's Artemis program?

- A) To send robots to explore Mars.
- B) To return astronauts to the moon for the first time since 1972.
- C) To develop new telescopes for deep space observation.
- D) To establish a permanent human colony on the moon by next year.